

Lab 3 SE

Architecture Selection & Justification

Project: Corporate Mental Wellness & Counseling Platform | **Domain:** Healthcare & Telemedicine

Architecture Selection

Chose Microservices Architecture for the Corporate Mental Wellness & Counseling Platform.

Reason 1 – Independent Scaling of Resource-Intensive Services

The platform's Video Consultation Service carries unpredictable, bandwidth-heavy load during peak counseling hours, while services such as Notification and Billing are comparatively lightweight. A microservices architecture lets each service scale independently, so compute resources can be directed to video sessions during peak demand without over-provisioning the entire system, unlike a layered or monolithic client-server design where all functionality scales together.

Reason 2 – Fault Isolation for a Multi-Actor, Compliance-Sensitive System

Corporate Employees and Licensed Therapists rely on different, loosely related workflows (scheduling, live video, clinical documentation, billing). Packaging these as independent services means a failure or update in one service, for example the Notification Service, cannot take down appointment booking or an active counseling session, which is essential in a healthcare context where availability directly affects patient care.

Security Advantage

Because each microservice owns its own data store and access boundary, the Clinical Records (EHR) Service, which holds protected health information (PHI) such as therapy session notes, can be isolated in a more tightly controlled network segment with stricter authentication than lower-sensitivity services. This limits the blast radius of a breach: compromise of a less sensitive service, such as Notification, does not expose PHI, which would be far harder to contain in a single shared layered database.

Performance Benefit

The Video Consultation Service can be deployed on infrastructure optimized for real-time audio/video (higher bandwidth, WebRTC-capable nodes) and auto-scaled independently during peak session hours, keeping call quality stable without slowing down lightweight, high-frequency operations such as appointment lookups or billing queries running on separate, right-sized services.

Components Identified (7)

API Gateway • User Management Service • Appointment Scheduling Service • Video Consultation Service • Clinical Records (EHR) Service • Notification Service • Billing & Insurance Service

Component Diagram:

